Mixture & Ratio – Solutions & Answers (Q21–Q40)

Detailed step-by-step solutions for questions 21 to 40.

Question 21

Pinky is in a queue. Ahead:Behind = 3:5. Total persons < 300. Maximum persons ahead of Pinky?

Correct Answer: 111

Solution:

- Let persons ahead = $3x$, behind = $5x$. Total = $3x + 5x + 1$ (Pinky) = $8x + 1$.
- Condition: $8x + 1 < 300 \rightarrow 8x < 299 \rightarrow x < 37.375$.
- Max integer $x = 37$. Max persons ahead = $3 \times 37 = 111$.

Question 22

Coffee jar: replace with cocoa twice. Coffee:Cocoa in Q = 16:9. Ratio of cocoa in P to cocoa in Q?

Correct Answer: C) 5:9

Solution:

- Let initial volume = V, amount replaced each time = x. After 2 replacements, coffee in Q = $V(1 - x/V)^2$.
- Coffee fraction in Q = 16/(16+9) = 16/25.
- $(1 - x/V)^2 = 16/25 \rightarrow 1 - x/V = 4/5 \rightarrow x/V = 1/5$.
- After 1 replacement (P): Coffee = $V \times 4/5$. Cocoa in P = V/5.
- Cocoa in Q = 9V/25.
- Ratio = (V/5) : (9V/25) = (5/25) : (9/25) = 5:9.

Question 23

Rama's score = (Mohan + Anjali)/12. After +6 each, revised ratio Anjali:Mohan:Rama = 11:10:3. Anjali exceeded Rama by?

Correct Answer: B) 32

Solution:

- Let revised scores: Anjali = $11x$, Mohan = $10x$, Rama = $3x$.
- Original scores: $a = 11x-6$, $m = 10x-6$, $r = 3x-6$.
- Initial relation: $r = (m+a)/12 \rightarrow 12r = m+a$.
- $12(3x-6) = (10x-6)+(11x-6) \rightarrow 36x-72 = 21x-12 \rightarrow 15x = 60 \rightarrow x = 4$.
- Anjali original = 44-6 = 38. Rama original = 12-6 = 6.
- Difference = 38 - 6 = 32.

Question 24

A:B = 1:3 by volume. Volume doubled by adding A → 72% alcohol. A = 60%. % alcohol in B?

Correct Answer: C) 92%

Solution:

- Initial mix: V of A and 3V of B. Total = 4V. Alcohol = 0.6V + 0.03bV (where b = % of B).
- Volume doubled by adding 4V of A. New total = 8V.
- Added alcohol from A = $0.6 \times 4V = 2.4V$.
- Total alcohol = (0.6 + 0.03b)V + 2.4V = V(3 + 0.03b).

• Final: $V(3 + 0.03b) = 0.72 \times 8V = 5.76V$.

• $3 + 0.03b = 5.76 \rightarrow 0.03b = 2.76 \rightarrow b = 92\%$.

Question 25

Bottle 1: milk:water = 7:2. Bottle 2: milk:water = 9:4. Combine for 3:1 milk:water. Ratio of volumes?

Correct Answer: B) 27:13

Solution:

• Milk proportion in B1 = 7/9, in B2 = 9/13, desired = 3/4.

• Apply alligation on milk fractions.

• Diff1 = 3/4 - 9/13 = (39-36)/52 = 3/52.

• Diff2 = 7/9 - 3/4 = (28-27)/36 = 1/36.

• Ratio = Diff1 : Diff2 = 3/52 : 1/36 = (3×36) : (1×52) = 108:52 = 27:13.

Question 26

a:b = 3:4 and b:c = 2:1. Which of 201, 205, 207, 210 is a possible value of (a+b+c)?

Correct Answer: C) 207

Solution:

• Combine: LCM of 4 and 2 is 4. b:c = 2:1 → 4:2. So a:b:c = 3:4:2.

• a+b+c = 9k for some positive integer k.

• Check options for divisibility by 9: 207 = 9 × 23. ✓

• 201 (digit sum 3), 205 (digit sum 7), 210 (digit sum 3) — not multiples of 9.

• Answer: 207.

Question 27

Mixture 1 (water:A = 1:2), Mixture 2 (water:B = 1:3), Mixture 3 (water:C = 1:4) mixed in 4:3:2. What is true about final mixture?

Correct Answer: C) More water than liquid B

Solution:

• Amounts taken: 4k, 3k, 2k.

• Water: (1/3)×4k + (1/4)×3k + (1/5)×2k = 4k/3 + 3k/4 + 2k/5 = 149k/60.

• Liquid A: (2/3)×4k = 8k/3 = 160k/60.

• Liquid B: (3/4)×3k = 9k/4 = 135k/60.

• Liquid C: (4/5)×2k = 8k/5 = 96k/60.

• Water (149) vs Liquid B (135): Water > Liquid B. ✓

• Water (149) vs Liquid A (160): Water < Liquid A. So (D) is false.

• Answer: More water than liquid B.

Question 28

Glass full of milk. 2/3 poured out and replaced with water, repeated 3 more times (4 total). Final milk:water?

Correct Answer: B) 1:80

Solution:

• Each time, 2/3 is removed so 1/3 milk remains.

• After 4 operations: milk = V × (1/3)■ = V/81.

• Water = V - V/81 = 80V/81.

• Ratio milk:water = (V/81) : (80V/81) = 1:80.

Question 29

Amal:Sunil = 3:2, Sunil:Mita = 4:5. Largest - smallest = Rs 400. Sunil's share?

Correct Answer: 800

Solution:

• Combined ratio: LCM of 2 and 4 is 4. A:S = 6:4, S:M = 4:5.

• A:S:M = 6:4:5. Largest = Amal (6k), smallest = Sunil (4k).

• Difference = 6k - 4k = 2k = 400 → k = 200.

• Sunil's share = 4k = 4 × 200 = Rs 800.

Question 30

187 fruits: apples:mangoes = 5:2. After selling 75 apples, 26 mangoes, half oranges, unsold apples:unsold oranges = 3:2. Total unsold fruits?

Correct Answer: 66

Solution:

• Let apples = 5x, mangoes = 2x. Oranges = 187 - 7x.

• Unsold apples = 5x-75, unsold oranges = (187-7x)/2.

• (5x-75)/((187-7x)/2) = 3/2 → 2(5x-75) = (3/2)(187-7x).

• 4(5x-75) = 3(187-7x) → 20x-300 = 561-21x → 41x = 861 → x = 21.

• Unsold apples = 30, mangoes = 16, oranges = 20. Total = 66.

Question 31

Grains sold to 3 customers: each gets half remaining + 3 kg. After 3rd customer, 0 remain. Initial quantity?

Correct Answer: C) 42 kg

Solution:

• Work backwards. Before 3rd customer: R■ - (R■/2 + 3) = 0 → R■ = 6 kg.

• Before 2nd customer: R■ - (R■/2 + 3) = 6 → R■/2 = 9 → R■ = 18 kg.

• Before 1st customer: x - (x/2 + 3) = 18 → x/2 = 21 → x = 42 kg.

Question 32

C1:C2:C3 profits = 9:10:8. C2:C4:C5 = 18:19:20. C5 profit = C1 profit + 19 crore. Total profit of all 5?

Correct Answer: A) 438 crore

Solution:

• Link ratios via C2: LCM(10,18) = 90. Multiply first ratio by 9, second by 5.

• C1:C2:C3 = 81:90:72. C2:C4:C5 = 90:95:100.

• Combined: C1:C2:C3:C4:C5 = 81:90:72:95:100.

• C5 - C1 = (100-81)k = 19k = 19 crore → k = 1 crore.

• Total = (81+90+72+95+100) × 1 = 438 crore.

Question 33

Tea A and B mixed and sold at Rs 40/kg. 10% profit when A:B = 3:2; 5% profit when A:B = 2:3. Cost prices ratio A:B?

Correct Answer: C) 19:24

Solution:

• Case 1 (3:2 mix): CP■ = 40/1.1 = 400/11. So (3C_A + 2C_B)/5 = 400/11 → 3C_A + 2C_B = 2000/11 × (11/11) = ... simplify: 33C_A + 22C_B = 2000.

• Case 2 (2:3 mix): CP■ = 40/1.05 = 800/21. So (2C_A + 3C_B)/5 = 800/21 → 2C_A + 3C_B = 4000/21. Multiply by 21: 42C_A + 63C_B = 4000 × (11/11)... simplify: 42C_A + 63C_B = 4000 × (1) = 4000 (scaled).

• From eq 1: multiply by 2: 66C_A + 44C_B = 4000.

• 66C_A + 44C_B = 42C_A + 63C_B → 24C_A = 19C_B → C_A/C_B = 19/24.

• Answer: 19:24.

Question 34

Popcorn packets L:S:J = 7:17:16. Chips L:S:J = 6:15:14. Total popcorn = total chips. Ratio of jumbo popcorn to jumbo chips?

Correct Answer: A) 1:1

Solution:

• Total popcorn = 7k+17k+16k = 40k. Jumbo popcorn = 16k.

• Total chips = 6m+15m+14m = 35m. Jumbo chips = 14m.

• Given 40k = 35m → 8k = 7m → k = 7x, m = 8x.

• Jumbo popcorn = 16×7x = 112x. Jumbo chips = 14×8x = 112x.

• Ratio = 112x:112x = 1:1.

Question 35

Amala, Bina, Gouri invest 3k:4k:5k at rates 6r:5r:4r. Bina's interest exceeds Amala's by Rs 250. Total interest?

Correct Answer: D) Rs 7250

Solution:

• Interest ∝ Principal × Rate (ignoring 1/100 factor).

• I_A = 3k×6r = 18kr; I_B = 4k×5r = 20kr; I_G = 5k×4r = 20kr.

• I_B - I_A = 20kr - 18kr = 2kr = 250 → kr = 125.

• Total = (18+20+20)kr = 58 × 125 = 7250.

Question 36

Vessels A (10%), B (22%), C (32%), each 500 ml. Transfer 100 ml A→B, then B→C, then C→A. Final strength of A?

Correct Answer: D) 14%

Solution:

• Initial salt: A=50g, B=110g, C=160g.

• Step 1 (A→B): Transfer 100ml (10g salt). A: 400ml, 40g. B: 600ml, 120g (strength=20%).

• Step 2 (B→C): Transfer 100ml from B (20g salt). B: 500ml, 100g. C: 600ml, 180g (strength=30%).

• Step 3 (C→A): Transfer 100ml from C (30g salt). C: 500ml, 150g. A: 500ml, 40+30=70g.

• Strength of A = (70/500) × 100 = 14%.

Question 37
Raju:Lalitha marbles = 4:9. After Lalitha gives some to Raju, ratio = 5:6. Fraction of Lalitha's original marbles given?

Correct Answer: D) 7/33

Solution:

• Let initial marbles: Raju = 4k, Lalitha = 9k. Total = 13k.

• After transfer, ratio = 5:6. Total still 13k. So new values: 5m and 6m with 11m = 13k.

• k=11, m=13. Raju: 44→65, Lalitha: 99→78. Marbles transferred = 65-44 = 21.

• Fraction = 21/99 = 7/33.

Question 38
Trader sells 10L of A+B mixture. B ≤ A. Cost A = Cost B + Rs 8. Sold for Rs 264 at 10% profit. Highest possible cost of B per litre?

Correct Answer: C) Rs 20

Solution:

• SP = 264. Profit = 10%. CP = 264/1.1 = 240.

• Let cost of B = C_B, cost of A = C_B + 8. Let a litres of A, b litres of B, a+b=10.

• Cost: a(C_B+8) + bC_B = 240 → 10C_B + 8a = 240 → C_B = (240-8a)/10 = 24 - 0.8a.

• Constraint: b ≤ a, i.e., 10-a ≤ a → a ≥ 5.

• To maximize C_B, minimize a. Min a = 5.

• Max C_B = 24 - 0.8×5 = 24 - 4 = Rs 20.

Question 39
Liquid 1: 1 kg/L, Liquid 2: 800 g/L. Half litre of mixture = 480 g. % of liquid 1 by volume?

Correct Answer: A) 80%

Solution:

• Density of mixture = 480g / 0.5L = 960 g/L.

• Apply alligation: D■=1000 g/L, D■=800 g/L, D_mix=960 g/L.

• Ratio V■:V■ = (960-800):(1000-960) = 160:40 = 4:1.

• % of liquid 1 = 4/(4+1) × 100 = 80%.

Question 40
Salaries Ramesh:Ganesh:Rajesh = 6:5:7 (2010), 3:4:3 (2015). Ramesh's salary +25%. % increase in Rajesh's salary?

Correct Answer: B) 7%

Solution:

• 2010 salaries: R=6x, G=5x, J=7x. 2015 salaries: R=3y, G=4y, J=3y.

• Ramesh 25% increase: 3y = 1.25×6x = 7.5x → y = 2.5x.

• Rajesh 2010 = 7x. Rajesh 2015 = 3y = 3×2.5x = 7.5x.

• % increase = (7.5x-7x)/7x × 100 = 0.5x/7x × 100 = 50/7 ≈ 7.14%.

• Closest to 7%.
